

Voice and Image Processing

2.1.4 Voice and image processing applications

2.1.4 Voice and Image Processing Applications (Detailed Explanation)

English + Nepali Mixup (No Romanize)

Introduction

Voice Processing र Image Processing दुवै Digital Signal Processing (DSP) का महत्वपूर्ण applications हुन्।

- **Voice Processing** मा speech वा audio signal लाई analyze, improve, compress, recognize, र transmit गरिन्छ।
- **Image Processing** मा digital image लाई analyze, enhance, restore, compress, र interpret गरिन्छ।

आजको समयमा AI, mobile phones, hospitals, CCTV, autonomous vehicles, satellite communication लगायत धेरै क्षेत्रमा Voice र Image Processing प्रयोग भइरहेको छ।

1. Voice Processing

Definition

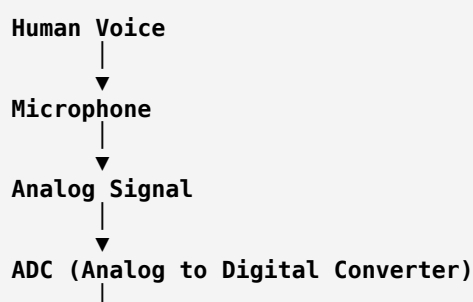
Voice Processing भनेको

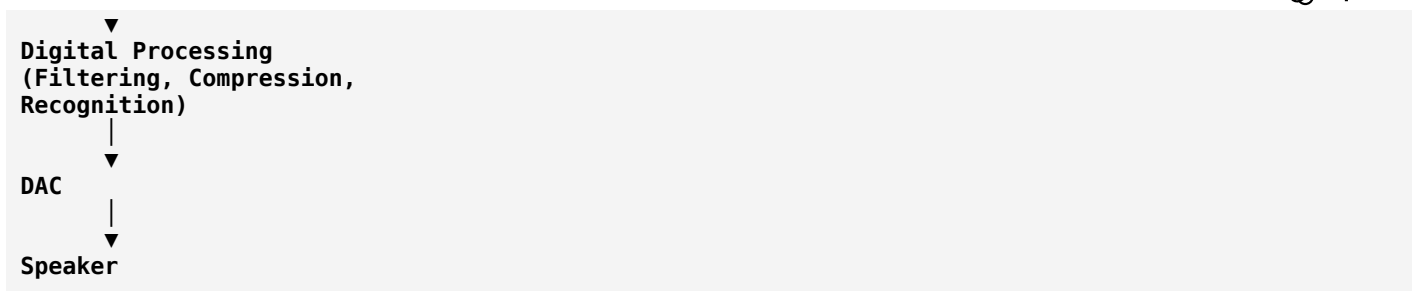
Human speech वा audio signal लाई digital form मा convert गरेर उसको quality improve गर्ने, recognize गर्ने, store गर्ने वा transmit गर्ने process हो।

Simple शब्दमा

Human Voice → Digital Signal → Processing → Better Output

Voice Processing Flow





Steps of Voice Processing

Step 1: Voice Acquisition

Microphone ले human voice capture गर्छ।

Example

"Hello"

यो analog signal हुन्छ।

Step 2: Sampling

Continuous signal लाई discrete samples मा divide गरिन्छ।

Example

Original Wave

~~~~~

↓

Samples

•  
•  
•  
•  
•

### Step 3: Quantization

Sample हरूलाई digital values मा convert गरिन्छ।

Example

Analog

2.34V

↓

Digital

10110101

### Step 4: Filtering

Noise हटाइन्छ।

Example

Before

Hello + Fan Noise

After

Clear Hello

## Step 5: Compression

Audio file को size घटाइन्छ।

Without compression

50 MB

After compression

5 MB

## Step 6: Recognition

Computer ले speech बुझ्छ।

Example

User:

Open Chrome

↓

Computer opens Chrome

# Applications of Voice Processing

## 1. Mobile Calling

Call गर्दा

- Noise reduction
- Echo cancellation
- Clear voice

DSP प्रयोग हुन्छ।

## 2. Voice Assistant

Examples

- Siri
- Google Assistant
- Alexa

Voice command recognize गर्छ।

Example

Turn on WiFi

## 3. Speech Recognition

Voice लाई text मा convert गर्छ।

Example

Speech

↓

"Hello Everyone"

↓

Text

## 4. Voice Authentication

Person को आवाजबाट identity verify गरिन्छ।

Example

Bank Voice Login

## 5. Online Meetings

Zoom

Google Meet

Microsoft Teams

Noise remove हुन्छ।

Echo cancel हुन्छ।

## 6. Hearing Aid

Old people को hearing device मा DSP प्रयोग हुन्छ।

Weak speech amplify हुन्छ।

Noise reduce हुन्छ।

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## 7. Music Applications

Spotify

YouTube Music

Equalizer

Bass Boost

Noise Reduction

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## 8. Medical Diagnosis

Doctor ले heart sound analyze गर्न सक्छ।

Lung sound analyze गर्न सक्छ।

Baby heartbeat monitor गर्न सक्छ।

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## Advantages of Voice Processing

- Clear audio quality
  - Noise reduction
  - Speech recognition
  - Fast communication
  - Secure authentication
  - Small storage requirement
  - Better user experience
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## Limitations

- Background noise ले error गर्न सक्छ।
  - Strong accent recognize गर्न गाह्रो हुन सक्छ।
  - High processing power चाहिन सक्छ।
  - Internet-based systems मा delay हुन सक्छ।
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## 2. Image Processing

### Definition

Image Processing भनेको

Digital image लाई analyze, modify, improve, restore, compress, वा understand गर्ने process हो।

Simple शब्दमा

**Image लाई computer प्रयोग गरेर better बनाउने process.**

### Image Processing Flow

Camera

↓

Image Capture

↓

Digital Image

↓

Image Processing

↓

Output Image

### Steps of Image Processing

#### Step 1: Image Acquisition

Camera ले image capture गर्छ।

Real Object

↓

Camera

↓

Image

#### Step 2: Preprocessing

Image improve गरिन्छ।

Examples

- Noise remove
- Brightness
- Contrast

## Step 3: Segmentation

Image लाई parts मा divide गरिन्छ।

Example

Face

↓

Eyes

↓

Nose

↓

Mouth

## Step 4: Feature Extraction

Important information निकालिन्छ।

Example

Face Detection

Fingerprints

Vehicle Number

Object Shape

## Step 5: Recognition

Computer ले object identify गर्छ।

Example

Cat

↓

Detected

↓

Cat

# Applications of Image Processing

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## 1. Medical Imaging

Hospital मा

- X-ray
- MRI
- CT Scan
- Ultrasound

Image enhancement गरिन्छ।

Doctor लाई diagnosis गर्न सजिलो हुन्छ।

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## 2. Face Recognition

Mobile Face Unlock

Attendance System

Airport Security

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## 3. CCTV Surveillance

Motion detection

Face detection

Vehicle tracking

Crime investigation

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## 4. Satellite Imaging

Weather forecast

Agriculture

Forest monitoring

Disaster management

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## 5. Autonomous Vehicles



Self-driving cars ले

- Road detect
  - Traffic signal detect
  - Pedestrian detect
  - Lane detect
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## 6. OCR (Optical Character Recognition)

Image मा भएको text लाई editable text मा convert गर्छ।

Example

Book Scan

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Editable Text

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## 7. Barcode and QR Code Scanner

Shopping Mall

Payment Apps

Inventory Management

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## 8. Photo Editing

Adobe Photoshop

Snapseed

Lightroom

Filters

Brightness

Contrast

Color correction

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## 9. Social Media

Instagram

Facebook

Snapchat

Automatic filters

Face beautification

Background blur

## 10. Industrial Inspection

Factory मा

Machine vision प्रयोग गरेर

- Crack detection
- Product quality inspection
- Defect detection

## Advantages of Image Processing

- Better image quality
- Fast analysis
- Automatic detection
- Medical diagnosis support
- High accuracy
- Reduced human effort
- Easy storage and sharing

## Limitations

- High computational cost
- Large storage requirement
- Poor image quality ले accuracy घट्न सक्छ।
- Complex algorithms आवश्यक पर्छन्।
- Lighting conditions ले performance प्रभावित गर्न सक्छ।

## Voice Processing vs Image Processing

| Voice Processing          | Image Processing           |
|---------------------------|----------------------------|
| Audio signal process गर्छ | Digital image process गर्छ |
| Input = Speech            | Input = Image              |

| Voice Processing        | Image Processing            |
|-------------------------|-----------------------------|
| Microphone प्रयोग हुन्छ | Camera/Scanner प्रयोग हुन्छ |
| Noise reduction         | Image enhancement           |
| Speech recognition      | Object recognition          |
| Voice assistant         | Face recognition            |
| Music processing        | Medical imaging             |

## Real-Life Example

### Smartphone Example

तपाईंले आफ्नो smartphone प्रयोग गर्दा एउटै device मा Voice Processing र Image Processing दुवै काम गरिरहेका हुन्छन्।

#### Voice Processing:

- "Hey Google" बोलेपछि phone ले तपाईंको आवाज recognize गर्छ।
- Call गर्दा background noise हटाएर clear voice दिन्छ।
- Voice typing गर्दा बोलेको कुरा text मा convert गर्छ।

#### Image Processing:

- Camera ले photo खिचेपछि brightness, contrast र color automatically सुधार गर्छ।
- Face Unlock ले तपाईंको अनुहार पहिचान गरेर phone unlock गर्छ।
- QR code scan गर्दा code लाई detect गरेर information निकाल्छ।

यसरी एउटै smartphone मा DSP प्रयोग गरेर voice र image दुवै efficiently process गरिन्छ।

## Examination Points (Important)

### Definition

- Voice Processing भनेको speech/audio signal लाई digital रूपमा process गर्ने प्रक्रिया हो।
- Image Processing भनेको digital image लाई improve, analyze, restore वा recognize गर्ने प्रक्रिया हो।

### Major Applications

- Voice Processing: Mobile calling, voice assistant, speech recognition, voice authentication, online meetings, hearing aids, music processing, medical diagnosis.
- Image Processing: Medical imaging, face recognition, CCTV surveillance, satellite imaging, autonomous vehicles, OCR, QR/barcode scanning, photo editing, industrial inspection.

### Key Benefits

- Voice Processing: Clear audio, noise reduction, speech recognition, efficient communication.

- **Image Processing:** Better image quality, automatic object detection, medical support, improved accuracy.

## Illustration

### Image Explanation:

1. **Voice Processing Pipeline** – Microphone → ADC → Digital Signal Processing → Speaker.
2. **Speech Processing** – Sampling, filtering, compression, and speech recognition.
3. **Image Processing Pipeline** – Camera → Image Acquisition → Preprocessing → Feature Extraction → Recognition → Output.
4. **Image Processing Applications** – Medical imaging, face recognition, CCTV surveillance, OCR, satellite imaging, and autonomous vehicles.

image illustration for pdf size

